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Intracranial pressure (ICP) is a vital determinant of brain function whose monitoring and treatment are central to the successful management of patients with a variety of brain diseases and disorders. There are also indications that changes in cerebral autoregulation - one of the primary physiologial mechanisms for maintaining normal ICP, and detectable via alteration of blood flow into the brain - can be altered during space flight in ways associated with space motion sickness and orthostatic intolerance. Currently, ICP is measured by drilling a hole into the skull and placing a device into one of several places within the cranium. Besides the risks attendant with this procedure, measurements of ICP can only be performed in hospitals staffed by neurosurgeons. This is clearly not an option available to astronauts in space, who often develop brain edema and its associated increase in ICP. Therefore, a simple, non-invasive method of monitoring ICP (nICP) would allow measurement of this critical parameter in astronauts. We have developed an algorithm for nICP based on novel analysis of standard diagnostic ultrasound of the brain and invasively and continually monitored arterial

Task Description:	blood pressure (ABP), and shown it feasible on 20 patients. We have also assembled and tested a device that allows collection of the data necessary to implement in real time, and to further refine this algorithm. Specific Aim #1: to test the feasibility of our approach to determining nICP on humans at several hospitals using invasive ABP (iABP), by collecting additional patient data to be used for refining the existing algorithm. Specific Aim #2: to perform the same feasibility tests and algorithm refinement with non-invasive ABP (nABP). Therefore, the work proposed here offers to perform steps necessary to bringing to space a non-invasive means of determining intracranial pressure. Key findings: in support of Aim #1, (a) we have automated the acquisition of data necessary to refine our nICP algorithm by consttructing of a web-based data entry portal that is also HIPPA compliant; (b) we have organized all necessary medical sites for our work, in conjunction with our industrial partner; (c) we have finished refinement of our prototype nICP device that also collects data necessary for finalizing the nICP algorithm; (d) we have designed and built a HIPPA compliant computer cluster necessary for processing the volumes of patient data we require for nICP algorithm development; (e) we have collected data from 49 patients using our prototype device and iABP at our Seattle hospital. In support of Aim #2, (a) we have identified two useful nABP devices; (b) we have shown that if their data quality is sufficient, that data from nABP will be sufficient to drive our nICP algorithm; (c) we have collected data from 6 patients using on nABP device at our Seattle hospital. Furthermore, we have preliminary laboratory data on a complementary method for using ultrasound to determine nICP. Finally, with our industrial partner we have made technical advances toward the development of a "user independent" transcranial Doppler system, which would allow the automated determination of nICP, and would also allow automated monitoring of cerebral blood flow and potential emboli into the brain within micro-g, high-g, and zero gravity environments. This automated Doppler system would be useful for studying and monitoring in situ, in an autonomous fashion, cerebral blood flow and cerebral autoregulatory contributions to neuro-vestibular alterations of function in general, and to crew performance, space motion sickness and orthostatic intolerance in particular. This system would also allow autonomous monitoring of emboli formation within the brain or movement of emboli from the peripheral vascular space to within the brain. We have contacted the "bed rest" researchers within NASA in order to lay the ground work necessary to test such a user independent system at NASA on volunteers with simulated micro-g physiology. Impacts: we have spent the first nine months of this study (all we've had available to us in the first year) laying the ground work for our proposed research, by establishing the medical sites, finishing our nICP prototype, and creating the necessary computational structures for data acquisition and processing. We have also demonstrated that we can collect useful data with our prototype system. These achievements have allowed us to move forward towards achieving the goals of our research project, namely to collect and analyze sufficient patient data to refine and finalize our approach to nICP. Finally, we have begun the work necessary to eventually introduce into NASA an autonomous transcranial Doppler system. In the second year of this project we will maintain our proposed pace of research. In particular, we will deploy our data acquisition and prototype nICP device at the remote sites in collaboration with our industrial partner; we will use the web-based data transfer system to bring that data to our home institution for analysis on our computer cluster; we will try out various nABP devices at our home institution within the context of our nICP research protocol; we will make further laboratory advances in the laboratory on a complementary means of non-invasively determining intracranial pressure; we will make further advances, resources permitting, for developing our user independent transcranial Doppler unit and attempt to deploy them within NASA's bed-rest test facilities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1) non-invasively predicting intracranial pressure (nICP): the patent literature and, to a lesser extent, the scientific literature, are littered with failed attempts to determine intracranial pressure in a non-invasive fashion. These considerable efforts have been moticated by the vital role played by this medical variable in the management of individuals suffering from head trauma, subarachnoid hemorrhage and other sources of intracranial bleeding, as well as brain tumors, hydrocephalus, edema, among other disorders. We have demonstrated in our preliminary work for this proposal a novel means of predicting nICP that has the best chance to date to capture this critical medical parameter whose current means of measurement requires that a neurosurgeon drill a hole through the skull. 2) In work with our industrial collaborators we have begun the earliest steps towards automating the transcranial Doppler process for the purposes of deriving medically useful information from measurements of blood flow rate the middle cerebral artery. Given such a system, this would allow researchers and clinicians to easily and autonomously measure the rate of blood flow into the brain - of intrinsic value and a critical parameter in our approach to the non-invasive determination of intracranial pressure. This system would also allow quantification of the existence and extent of emboli transport into the brain, and the quality of cerebral autoregulation. The monitoring of these measurements plays an important role in guiding the medical and/or surgical care of patients with a variety of neurological disorders (as listed above). 3) Others have noted in the literature that transcranial Doppler ultrasound can, in conjunction with various pharmaceutical agents, can accelerate the destruction of unwanted blood clots in the brain - the major cause of stroke for several hundred thousand individuals in the USA every year. An autonomous transcranial Doppler system focused on the middle cerebral artery would allow the rapid and automatic focus of ultrasound onto the site of the majority of blood clots within the brain, making ultrasound-assisted thrombolysis readily deployable in the field.
Task Progress:	We have established the computational and logistical infrastructure necessary to most effectively pursue our human-based studies on the non-invasive determination of intracranial pressure. Regarding computational infrastructure, we have built a computer cluster of sufficient power and connectivity to efficiently run through what we anticipate will be terabytes worth of human physiological data through our mathematical formalisms. Regarding logistical infrastructure, we have completed a web-based data entry system that will allow all of our clinical sites to transmit the human data they collect in a HIPPA compliant way to the University of Washington for analysis. We have also determined in large degree the complexity of the particular mathematical model we are using to relate in an empirical fashion readily observable human physiological data to intracranial pressure.

	Finally, we have finished altering our prototype non-invasive ICP device. We have deployed it at Harborview Medical Center in Seattle, where we have collected human data that demonstrate the usefulness of the device. This human physiological data also directly supports of our research effort effort toward the non-invasive determination of ICP. We are therefore ready to deploy our device to our remote clinical sites. With these findings we are ready to collect the bulk of the human data we need to finalize our intracranial pressure algorithm.
Bibliography Type:	Description: (Last Updated: 11/21/2013)
Awards	"For Dan Blizzard, an undergraduate reseacher: NIH Clinical Research Experience for Engineers." Jan-2006
Awards	"For Dan Blizzard, an undergraduate reseacher: Washington Space Grant NASA Undergraduate Research Award." Jan-2006
Awards	"for Dan Blizzard, an undergraduate reseacher: Washington Space Grant NASA Summer Undergraduate Research Program." Jan-2006
Awards	"for Pavan Vaswani, an undergraduate researcher: Washington Space Grant NASA Summer Undergraduate Research Program." Jan-2006